



Energy Management System

brodotehna
CONTROL SYSTEMS



SYSTEM OVERVIEW

Energy Management System (aEMS)



Advanced Energy Management System (aEMS) manages energy production, consumption and storage at the customer site. It can be deployed on-site or in the cloud (cloud-based).

AEMS is essential in systems that include a **Battery Energy Storage System (BESS)**, as it enables optimal battery operation using **advanced forecasting (AI)** and **mathematical optimization** to achieve the customer's operational and economic objectives.

The system supports multiple operating modes, including:



1. Production/consumption schedule mode



2. Cost minimization mode

PRODUCTION MODE

Production/Consumption Schedule Mode

This mode is used when the **user wants to follow a predefined energy exchange schedule with the power grid or energy market.**

In this operating mode, the system:



Tracks the **defined reference power** at the point of connection



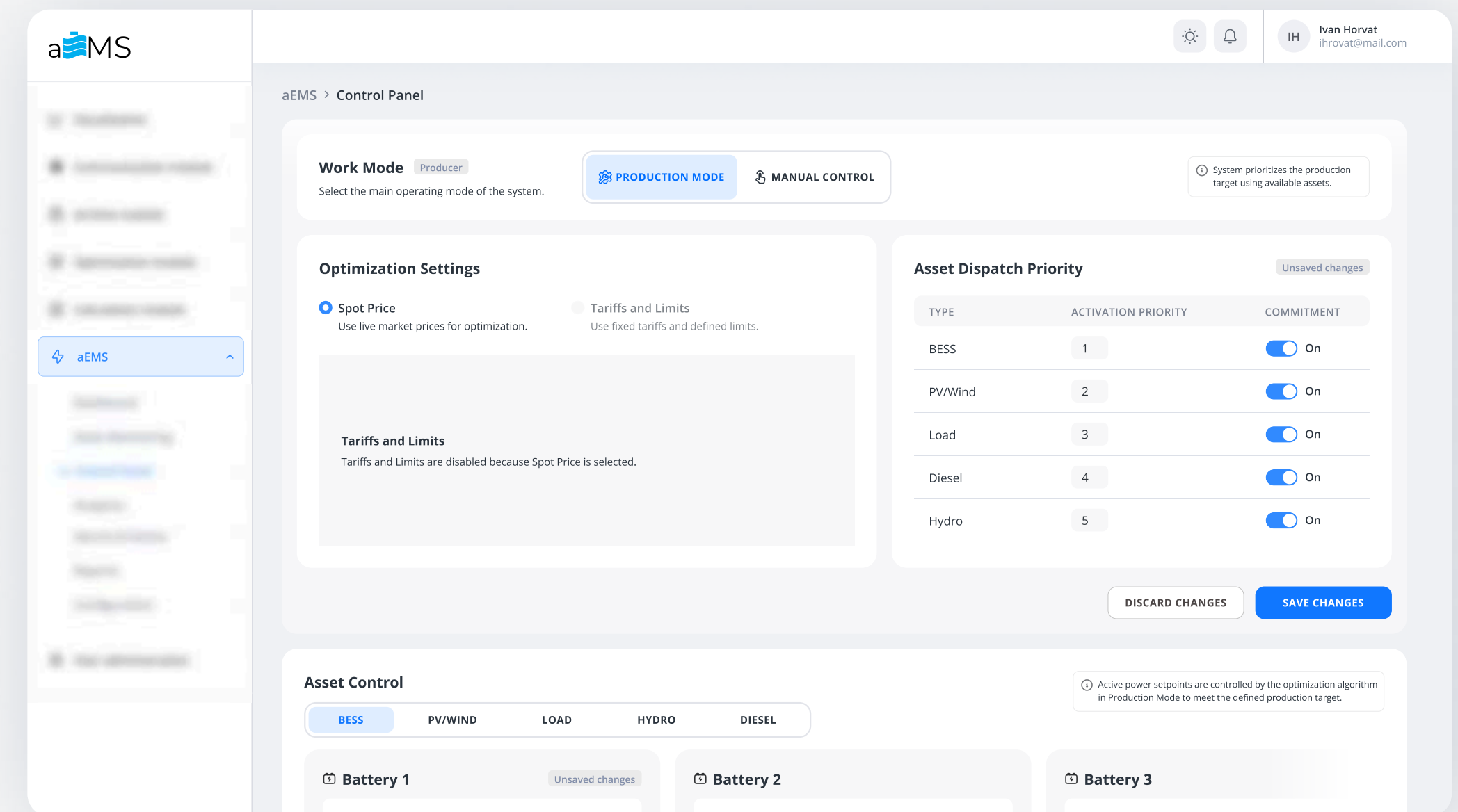
Controls **flexible resources** to achieve the planned production or consumption profile



Compensates for deviations caused by **changes in production or consumption**



Generates **market-optimized production and exchange schedules** and imports reference schedules via the **IEC 104 protocol**



PRODUCTION MODE

Production/Consumption Schedule Mode

PARTICULARLY SUITABLE FOR:



Users with dominant energy
production
(e.g. PV or hybrid power plants)



Participation in the
electricity market



Integration with Virtual Power Plants
(VPPs)

COST MINIMIZATION

Energy Cost Minimization Mode

The energy cost minimization mode is used when the primary objective is **to reduce total electricity costs**, including the cost of **imported energy and peak demand** within a given billing period, for example by optimizing **grid charges and energy costs**.

In this operating mode, the system:



Continuously monitors **power at the point of connection**



Forecasts future **consumption and production**



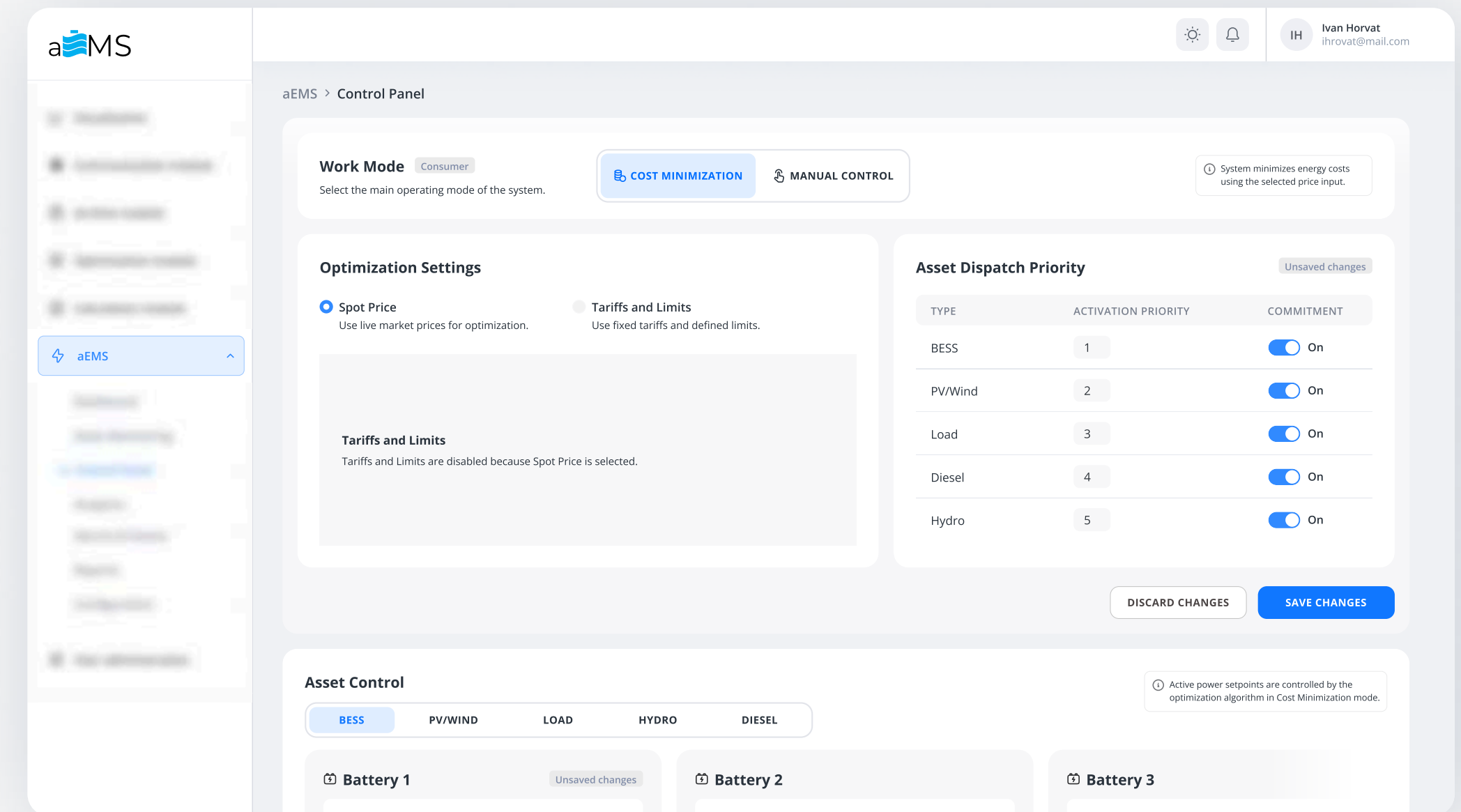
Controls the **battery and other flexible resources** to limit peak demand



Keeps consumption **within defined limits**



Uses configured **tariffs or electricity market prices**



COST MINIMIZATION

Energy Cost Minimization Mode

PARTICULARLY SUITABLE FOR:



Industrial and commercial consumers



Optimization of peak demand and
energy consumption charges



Protection of grid and
transformer capacity

SYSTEM CAPABILITIES

Software Capabilities:

→ Battery Energy Storage System (BESS) Management



Battery charging and discharging control



State of charge (SOC) limit protection



Charging and discharging power limits



Battery operation optimization for cost reduction or market optimization



SYSTEM CAPABILITIES

Software Capabilities:

→ Generation Source Management



Monitoring renewable energy generation (PV, wind)



Optional generation curtailment



Integration of generation into the control algorithm



SYSTEM CAPABILITIES

Software Capabilities:

→ Load Management



Monitoring total and individual consumption



Integration of controllable loads

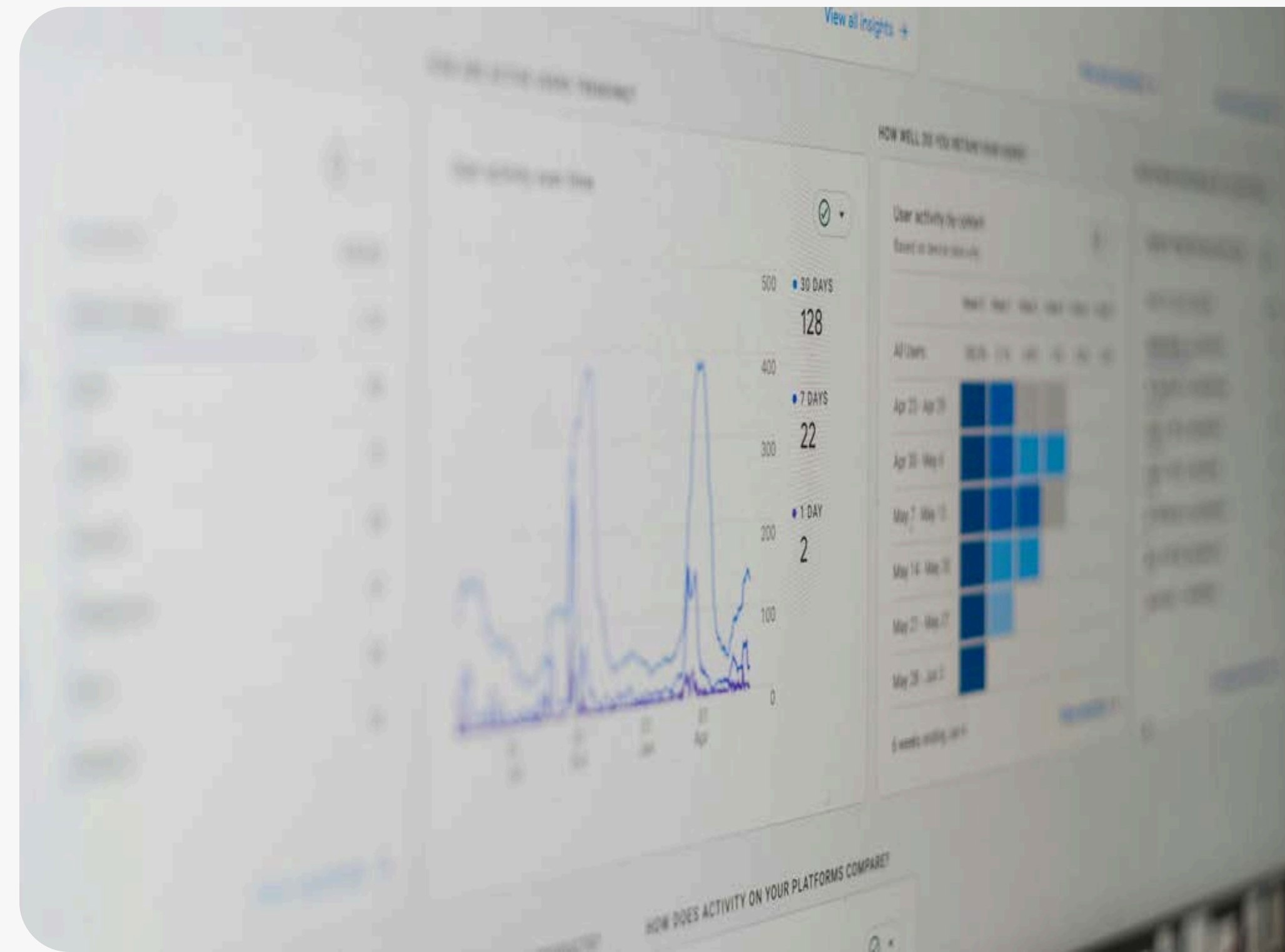
→ Power Management at the Point of Connection



Continuous monitoring of power exchange with the grid



Detection of defined limit violations



SYSTEM CAPABILITIES

Software Capabilities:

→ AI Capabilities

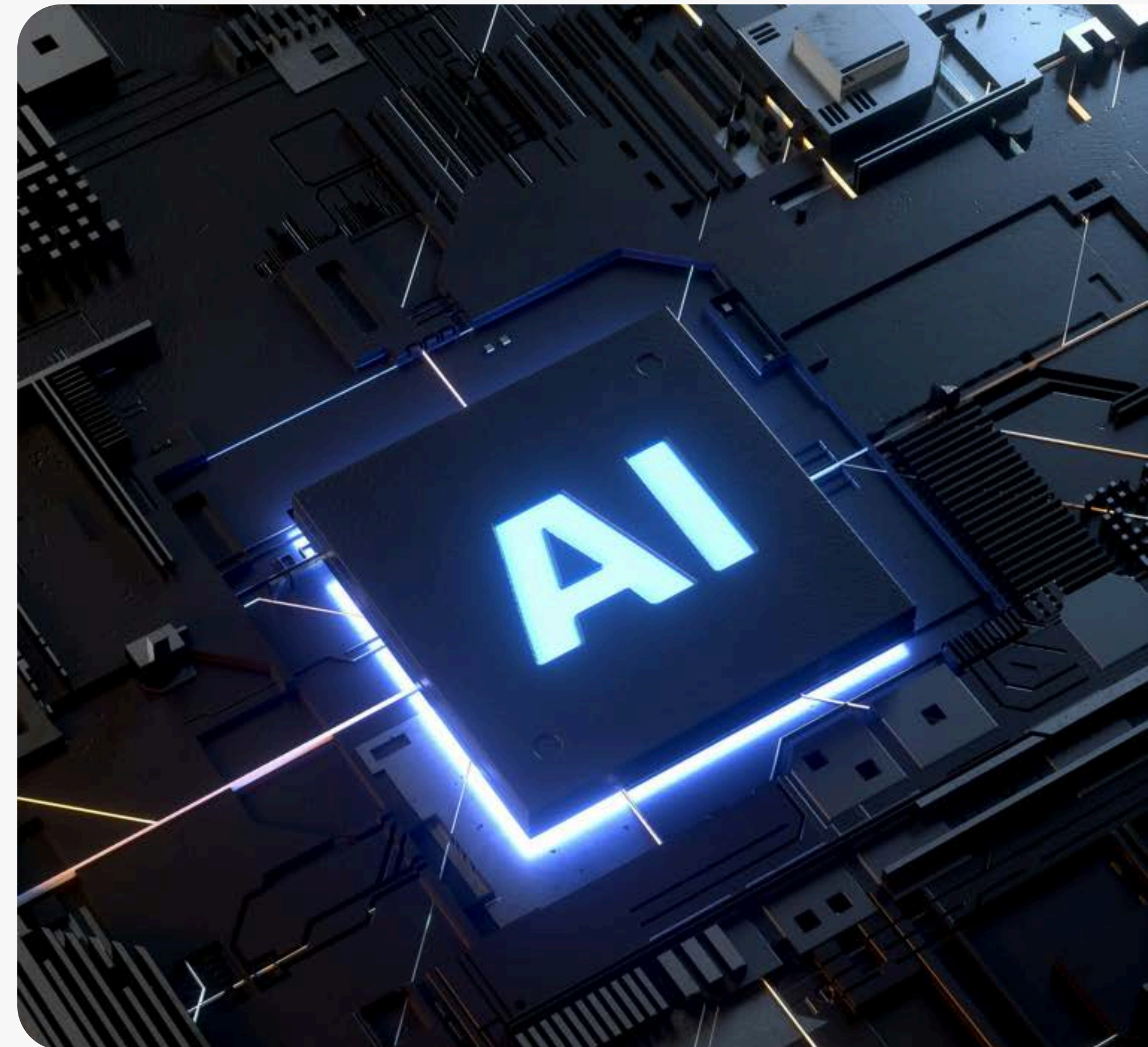
The system uses artificial intelligence (AI) to support control decisions.

Supported forecasts:

- Energy consumption
- Renewable energy generation

Based on these forecasts, the system:

- Predicts future peak demand
- Estimates the energy required for peak shaving
- Plans battery operation



MONITORING & VISUALIZATION

Visualization and User Interface

The system includes a **web-based user interface** for monitoring and control.

Visualization includes:

- Current consumption, generation and power at the point of connection
- Battery status (SOC, power, energy)
- Energy flows within the system
- Active operating mode (schedule-based mode / cost minimization mode)

Graphical displays provide:

- Historical data overview
- Forecast visualization
- Comparison of planned and actual energy flows

Asset Control

BESS

PV/WIND

LOAD

HYDRO

Battery 1

Unsaved changes

Active Power Setpoint

Current Value 9.70

New Value

Min SOC Limit

75 kW

Max SOC Limit

188 kW

Activation Priority 1

Commitment ☒ On

MANAGEMENT & CONFIGURATION

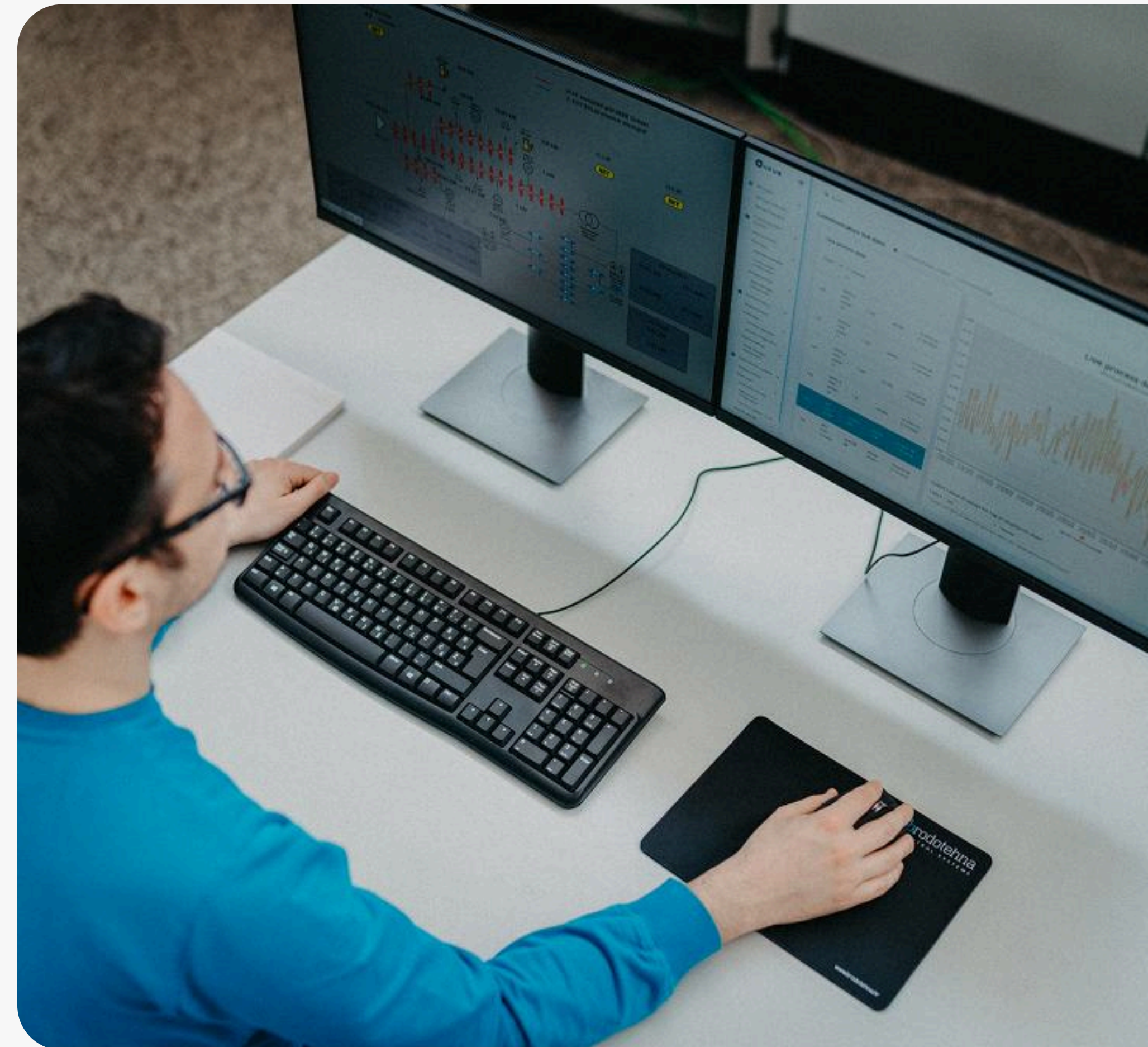
System Settings

The system allows configuration of key parameters.

Settings include:

- Configuration of tariff options or electricity market prices
- Power limits at the point of connection
- Battery parameters
- Control algorithm configuration
- Definition of system safety limits

Configuration can be performed through the web-based user interface with administrator access.



SYSTEM SAFETY

Monitoring and Alarming

The system includes monitoring and alarm functionalities.

Supported functions:

✓ Alarm generation when defined limits are exceeded

✓ Monitoring the status of connected devices



SYSTEM CONNECTIVITY

Integration and Communication

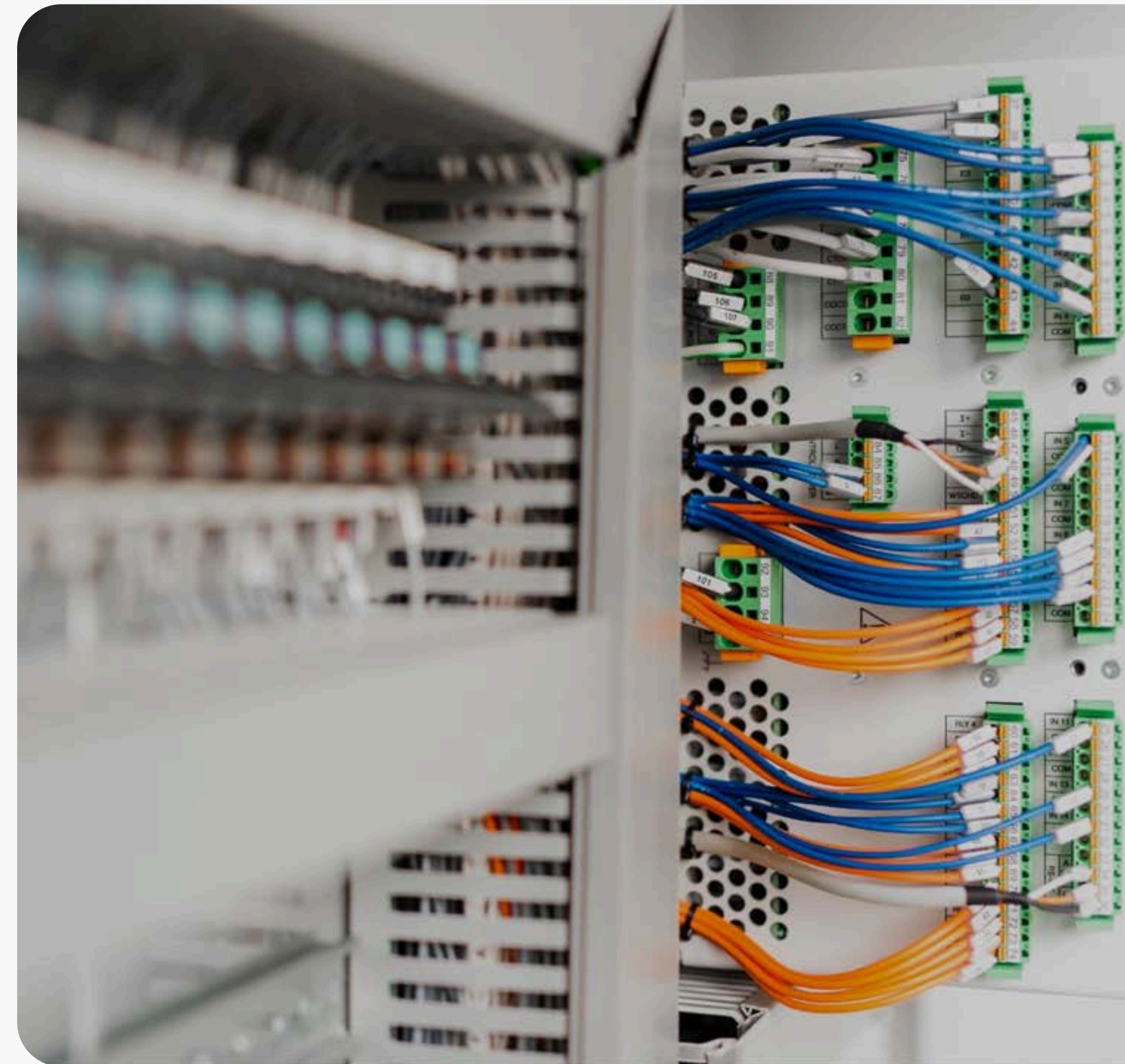
The system enables integration with other systems through standard communication protocols.

Supported protocols may include:

- Modbus TCP client
- IEC 61850 client
- IEC 60870-5-104 client/server
- S7 Siemens client
- REST API (CROPEX)

This enables integration with:

- SCADA systems
- Energy market platforms for day-ahead price retrieval
- Virtual Power Plant (VPP) systems



Contact Information



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Thank you!

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